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COMMON QUARTERLY EXAMINATION – 2025

XII SET2

Standard XII

Reg.No.:

PHYSICS

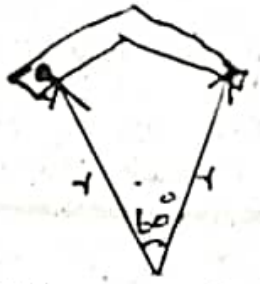
Time: 3.00 hrs.

Marks: 70

Part - I

I. Choose the correct answer: 15 x 1 = 15

1. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If those are connected by a conducting wire, the final charge on the sphere is
   a) 1×10^{-2} C b) 3×10^{-2} C c) 2×10^{-2} C d) 4×10^{-2} C
2. Two identical conducting balls having positive charges q_1 and q_2 are separated by a center to center distance 'r'. If they are made to touch each other and then separated to the same distance, the force between them will be
   a) more than before b) less than before c) zero d) same as before
3. A toaster operating at 240 V has resistance of 120 Ω. Its power is
   a) 240 W b) 480 W c) 2 W d) 400 W
4. Two wires of A and B with circular cross section are made up of the same material with equal lengths, suppose $R_A = 3R_B$, then what is the ratio of radius of wire A and B?
   a) 3 b) $\sqrt{3}$ c) $\frac{1}{3}$ d) $\frac{1}{\sqrt{3}}$
5. A bar magnet of length 'l' and magnetic moment P_m is bent in the form of an arc as shown in figure. The new magnet dipole moment will be



   a) P_m b) $\frac{2}{\pi}P_m$
   c) $\frac{1}{2}P_m$ d) $\frac{3}{\pi}P_m$
6. A wire of length 'l' carrying a current I along the y direction is kept in a magnetic field given by $\vec{B} = \frac{\beta}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$. The magnitude of Lorentz force acting on the wire is
   a) $\sqrt{2}\beta Il$ b) $\sqrt{\frac{2}{3}}\beta Il$ c) $\sqrt{\frac{1}{2}}\beta Il$ d) $\sqrt{\frac{1}{3}}\beta Il$
7. In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current in the circuit is
   a) $\pi/6$ b) $\pi/4$ c) zero d) $\pi/2$
8. The flux linked with a coil at any instant t is given by $\phi_B = 15t^2 - 50t + 250$. The induced emf at t = 3s is
   a) –40 V b) –190 V c) 40 V d) –10 V
9. If a material having intensity of magnetisation 500 Am^{-1} is placed in a magnetising field of 1000 AM^{-1}, then the susceptibility of the material is
   a) 0.2 b) 0.8 c) 0.7 d) 0.5
10. An example of diamagnetic material is
    a) Nickel b) Water c) Aluminium d) Iron
11. If the magnitude of the magnetic field is 3×10^{-6} T, then the magnitude of the electric field for a electromagnetic wave is
    a) 600 Vm^{-1} b) 100 Vm^{-1} c) 900 Vm^{-1} d) 300 Vm^{-1}
12. The speed of light in an isotropic medium depends on
    a) its intensity b) its wavelength
    c) the nature of propagation d) the motion of the source w.r.t. medium

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13. Stars twinkle due to
a) reflection b) total internal reflection c) refraction d) polarisation
14. For the light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
a) 30° b) 45° c) 60° d) 90°
15. A plane glass is placed over a various coloured letters. The letter which appears to be raised more is
a) red b) yellow c) green d) violet

Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Define electric field.
17. State the applications of Seebeck effect. (Any two)
18. How will you increase the current sensitivity of a galvanometer?
19. Explain the reason for glittering of diamond.
20. The ratio of intensities of two waves in an interference pattern is 36:1. What is the ratio of the amplitudes of the two interfering waves?
21. Why electro magnetic waves are non mechanical?
22. What is meant by hysteresis?
23. How will you define Q-factor?
24. The angle of minimum deviation for the equilateral power is 40°. Find the refractive index of the material of the prism.

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Obtain a relation between current and drift velocity.
26. Derive the relation between f and R for a spherical mirror.
27. Write the uses of microwaves.
28. Explain the conversion of galvanometer into an ammeter.
29. Derive an expression for electrostatic potential due to a point charge.
30. Explain the various energy losses in transformer.
31. State Huygen's principle.
32. The resistance of nichrome wire at 20°C is 10Ω . If it's temperature coefficient of resistance is 0.004/°C. Find its resistance at boiling point of water. Comment on the result.
33. Find the impedance of a series RLC circuit, if the inductive reactance, capacitive reactance and resistance are 184Ω, 144Ω and 30Ω respectively. Also calculate the phase angle between voltage and current.

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) i) State Coulomb's law in electrostatics.
ii) State the differences between Coulomb force and gravitational force. ***(OR)***
b) Derive an expression for electrostatic potential due to an electric dipole.
35. a) Explain the determination of unknown resistance using metre bridge. ***(OR)***
b) How the emf of two cells are compared using potentiometer?
36. a) Discuss the working of cyclotron in detail. ***(OR)***
b) Using Biot-Savart law deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current.
37. a) Explain the construction and working of transformer. ***(OR)***
b) Obtain the equation for bandwidth in Young's double slit experiment.
38. a) What is absorption spectrum? Explain the types of absorption spectrum? ***(OR)***
b) Derive the mirror equation and the equation for lateral magnification.

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